

EXPERIMENT - 16

1's complement.

AIM:

To write assembly language to find 1's complement by using 8085 microprocessor simulator.

ALGORITHM:

- 1) loads the value from memory address 8000 into A accumulator.
- 2) complements the bits of the value in A using CMA (complement accumulator) instruction.

This means that all 1's become 0s and all 0s becomes 1s.

- 3) stores the complemented value into memory address 8001.

- 4) Halts the program execution.

PROGRAM:

MNEMONICS	EXPLANATION
LDA 8000	load the contents of memory address 8000 into Accumulator

CMA

Complement the contents of accumulator - or A

STA 8001

store the complemented value into memory address 8001

HLT

Halt the program execution.

INPUT:

Address	Data
8000	6

OUTPUT:

ADDRESS	DATA
8001	249

0184

RESULT:

This program was executed successfully by using 8085 simulator.



Registers			Flag	
A	F9		S	0
BC	00	00	Z	0
DE	00	00	AC	0
HL	00	00	P	0
PSW	00	00	C	0
PC	42	08		
SP	FF	FF		
Int-Reg	00			

Decimal - Hex Conversion

Decimal	Hex
0	0
To Hex	To Dec

I/O Ports

0	-	+	00
Update Port Value			

Memory

8000	-	+	06
Update Memory			

Load me at

```
1 LDA 8000
2 CMA
3 STA 8001
4 HLT
```

Data Stack KeyPad **Memory** I/O Ports

Start

Address (Hex)	Address	Data
1F41	8001	249
1F42	8002	0
1F43	8003	0
1F44	8004	0
1F45	8005	0
1F46	8006	0
1F47	8007	0
1F48	8008	0
1F49	8009	0
1F4A	8010	0
1F4B	8011	0
1F4C	8012	0
1F4D	8013	0
1F4E	8014	0
1F4F	8015	0
1F50	8016	0

Line No	Assembler Message
0	Program assembled successfully